CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 27: Graphical Solution

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Write three points on the line $x + 2y = 6$ suitable for plotting.
2. Solve graphically (or verify): $x + y = 7$ and $x - y = 1$.
3. Solve: $2x + y = 8$ and $x - y = 1$.
4. Solve: $y = 2x$ and $x + y = 6$.
5. How are the lines $3x - y = 2$ and $6x - 2y = 7$ related?
6. How are the lines $x + y = 3$ and $2x + 2y = 6$ related?
7. For what value of k are $2x + ky = 5$ and $4x + 6y = 1$ parallel?
8. The lines $x + y = 6$ and $x - y = 2$ meet at $(4, 2)$. Find the area of the triangle they form with the x-axis.
9. Find the area of the triangle formed by the lines in the previous question and the y-axis.
10. Find the vertices of the triangle formed by $y = x$, $y = 0$ and $x = 4$, and its area.

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Answer Key

1. For example $(0, 3)$, $(2, 2)$, $(6, 0)$
2. $x = 4, y = 3$
3. $x = 3, y = 2$
4. $x = 2, y = 4$
5. Parallel; no solution
6. Coincident; infinitely many solutions
7. $k = 3$
8. Base from $(2, 0)$ to $(6, 0)$ is 4; height 2; area 4 square units
9. Base from $(0, -2)$ to $(0, 6)$ is 8; height 4; area 16 square units
10. $(0, 0)$, $(4, 0)$, $(4, 4)$; 8 square units

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